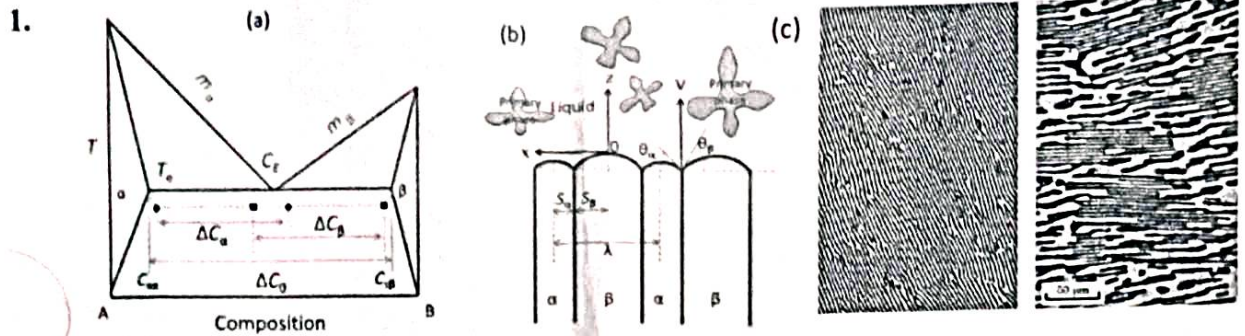


End Term

COURSE: Thermodynamics of Materials

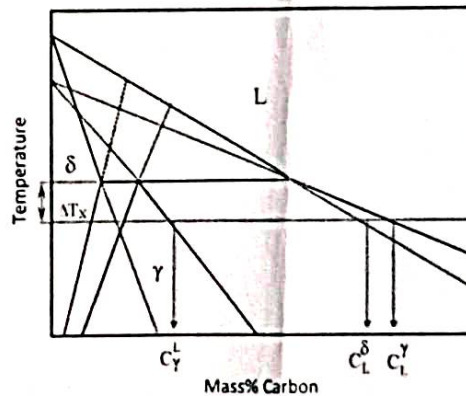
Total time: 60 mins

Answer All the Questions (Total = 30)



- (a) Develop a suitable equation with associated schematic to describe the evolution and related growth of the Microstructure as shown above. How can you control the λ in this case. Mathematically show the effect of Temperature on this growth model. In which condition, the concentration gradient will vanish. (5+2+2+1=10)

2.



Briefly discuss what type of Industrial Manufacturing Challenges can be solved by the diagram shown above (use schematic if required). Explain the diagram accordingly.

(5+5=10)

3. Consider a Ternary Alloy System of A, B and C element. The regular solution model for a ternary alloy system could be expressed as follows (Notations have their standard meaning and all the Ω are positive in nature):

$$G^{\alpha} = {}^{\circ}G_A^{\alpha} X_A + {}^{\circ}G_B^{\alpha} X_B + {}^{\circ}G_C^{\alpha} X_C + \Omega_{AB}^{\alpha} X_A X_B + \Omega_{BC}^{\alpha} X_B X_C + \Omega_{CA}^{\alpha} X_A X_C + RT(X_A \ln X_A + X_B \ln X_B + X_C \ln X_C)$$

Develop an expression of Miscibility Gap in terms of Temperature and Composition for this system. Draw the schematic Miscibility Domain in the T-X diagram. (10)

MT-2-04(TO) - Thermodynamics of Materials

End-Semester Exam
Date: 26th November 2025

Total Marks [10/40]
Time: 14:30 to 17:30 hrs

PART-B

B1. Compare Ellingham diagram with Pourbaix diagram for following aspects:

- | | |
|---|------|
| (i) Ordinate and Abscissa | [01] |
| (ii) Slope and Intercept | [01] |
| (iii) Case(s) when horizontal line and vertical line will appear with example(s) if any | [02] |
| (iv) Case(s) which implies to increase and decrease in slope | [01] |

B3. For a hypothetical metal oxide M_2O_3 , polyatomic anion MO_2^- and metal M, consider the thermodynamic data for following reactions as:

- $\Delta G^\circ (M_2O_3 \rightleftharpoons M) = 10 \times \alpha \text{ kJ}$,
- $\Delta G^\circ (MO_2^- \rightleftharpoons M) = 9 \times \alpha \text{ kJ}$,
- $\Delta G^\circ (M_2O_3 \rightleftharpoons MO_2^-) = 5 \times \alpha \text{ kJ}$,

consider α = last two digits of your student ID, Temperature 298 K, and concentration of $[MO_2^-] = 1.0$

- (i) Derive the equations E as function of pH using values of ΔG° for each of the three reactions, after balancing using the proper procedure
- (ii) Plot the equation derived using graph sheet to generate the Pourbaix diagram (E vs. pH) of this hypothetical metal to metal oxide.
- Mention appropriate scales, axes labels and units in the graph sheet*

[03] 2.5

[03] 2.5